

■ MODEL ANSWERS: Mathematics Paper 2: Reasoning

Step-by-Step Solutions for Ayaan

A: $38 - w - 5$

Method: Step 1: Start with total matches (38). Step 2: Subtract wins (w) and losses (5). The remaining matches are draws. Expression: $38 - w - 5$ (or $33 - w$).

A: 25 emeralds

Method: Step 1: Ratio is 3:5. Step 2: Ayaan found 15 diamonds. 15 is 3×5 . Step 3: Multiply the emerald part by the same number (5): $5 \times 5 = 25$.

A: 3,862.4m (approx) or 3,840m if using 1.6km conversion

Method: Step 1: 1 mile is approx 1.6km. $2.4 \text{ miles} \times 1.6 = 3.84\text{km}$. Step 2: Convert km to m by multiplying by 1000. $3.84 \times 1000 = 3,840\text{m}$.

A: 25

Method: Step 1: To find x, subtract 15 from both sides of the equation. Step 2: $40 - 15 = 25$. So, $x = 25$.